Learning with Class-Conditional Label Noise: Forward Correction, Generalised Cross-Entropy and Anchor-Point Transition Estimation

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Abstract

Deep classifiers tend to memorise wrong labels, so label noise directly harms their accuracy. This report studies three-class image classification in which the training and validation labels are corrupted by class-conditional noise while the test labels are clean. We compare two noise-robust classifiers: *forward loss correction*, which uses the noise transition matrix $\boldsymbol{T}$, and the *generalised cross-entropy* (GCE) loss, a bounded robust loss that does not need $\boldsymbol{T}$ and was not covered in the course. When $\boldsymbol{T}$ is unknown, we estimate it with an *anchor-point estimator* and validate it on the two FashionMNIST datasets, where $\boldsymbol{T}$ is given. Over ten random data splits, forward correction is more accurate on average on all three datasets, whereas GCE is far more stable on CIFAR, where forward correction with the estimated matrix occasionally converges to a class-permuted solution. We also analyse why model selection on noisy validation data is difficult under heavy noise. Section 1 introduces the problem, Section 2 reviews related work, Section 3 describes the methods, Section 4 reports and discusses the experiments, Section 5 concludes, and Appendix A explains how to run the code.

1 Introduction

Supervised learning assumes correct labels. In practice, large datasets are labelled by crowd-sourcing, by web search keywords or by automatic tools, and annotators confuse similar classes. Label noise is therefore the normal condition for real-world data [1]. It is especially harmful for deep networks. Over-parameterised networks can fit completely random labels [2]. They first learn simple patterns and then memorise the mislabelled examples [3], so standard cross-entropy (CE) training

generalises poorly. Learning robustly from noisy labels makes cheap annotation usable and makes systems trained on web-scale, crowd-sourced or medical data, where even experts disagree, more reliable.

We consider *class-conditional noise* (CCN). The observed label $\tilde{Y}$ depends on the image only through its clean label Y: $P(\tilde{Y}{=}j \mid Y{=}i, X) = T_{ij}$, where $\boldsymbol{T}$ is the row-stochastic *transition matrix* [4–6]. The task has two parts: (i) train classifiers on noisy data that are accurate on clean test data when $\boldsymbol{T}$ is known; (ii) estimate $\boldsymbol{T}$ when it is unknown and use the estimate for classification. We use **forward correction** [6], which fits $\boldsymbol{T}$ times the predicted clean posterior to the noisy labels. Our second classifier uses the **generalised cross-entropy (GCE)** loss [7], a bounded loss that interpolates between CE and the mean absolute error and needs no knowledge of $\boldsymbol{T}$. Comparing the two shows how much is gained by modelling the noise explicitly. For the unknown $\boldsymbol{T}$ we use an **anchor-point estimator** [5, 6], which we first check against the two known matrices.

2 Related work

Label-noise methods fall into four families [1]. Table 1 lists their advantages and disadvantages. **Loss correction** uses the relation $P(\tilde{Y} \mid x) = \boldsymbol{T}^{\top} P(Y \mid x)$. Natarajan et al. [4] build an unbiased estimator of the clean loss for binary problems. Liu and Tao [5] treat the noisy data as a shifted distribution and reweight each example by $\beta = P(Y{=}\tilde{y} \mid x)/P(\tilde{Y}{=}\tilde{y} \mid x)$. Patrini et al. [6] proposed *backward* correction (the loss vector multiplied by $\boldsymbol{T}^{-1}$) and *forward* correction (the predicted posterior multiplied by $\boldsymbol{T}$) for deep networks. With the true $\boldsymbol{T}$ these methods are statistically consistent: minimising the corrected noisy risk yields the clean Bayes classifier. **Transition-matrix estimation** mostly relies on *anchor points*, i.e. instances that belong to one class with certainty, whose noisy posterior equals a row of $\boldsymbol{T}$ [5]. The anchors are approximated by the points with the largest estimated noisy posterior [6]. Dual-T [8] reduces the estimation error. T-Revision [9] avoids the need for exact anchor points by refining $\boldsymbol{T}$ during training. **Robust losses** need no $\boldsymbol{T}$. Symmetric losses such as MAE tolerate symmetric noise but underfit deep networks [10]. GCE [7] trades this robustness against ease of fitting. **Sample selection and regularisation** methods, such as Co-teaching [11], exploit the fact that networks fit clean labels first. They treat small-loss samples as clean or regularise towards the early predictions.

3 Methods

3.1 Problem formulation

Let x be an image, $Y \in \{0, 1, 2\}$ its clean label and $\tilde{Y}$ its noisy label. The training data $\{(x_n, \tilde{y}_n)\}_{n=1}^{N}$ are drawn from the noisy distribution D_ρ; evaluation uses the clean distribution D. A network f_θ outputs logits, and $\boldsymbol{p}(x) = \mathrm{softmax}(f_\theta(x))$ estimates $P(Y \mid x)$. We store $\boldsymbol{T}$ row-wise, $T_{ij} = P(\tilde{Y}{=}j \mid Y{=}i)$, which is the transpose of the column convention used in the lectures. Under CCN the sum rule

Table 1: Main families of label-noise methods: advantages and disadvantages.

Family	Advantages	Disadvantages
Loss correction [4–6]	Statistically consistent; model-agnostic; theoretically grounded	Needs an accurate $\boldsymbol{T}$; backward correction gives negative, high-variance losses if $\boldsymbol{T}$ is ill-conditioned
$\boldsymbol{T}$ estimation [6, 8, 9]	Makes correction possible when $\boldsymbol{T}$ is unknown; anchor method is simple	Needs anchor or identifiability assumptions; biased by errors in the estimated posterior
Robust losses [7, 10]	No $\boldsymbol{T}$ needed; one-line change to training	Weaker guarantees under asymmetric noise; MAE-like losses underfit
Sample selection [11]	Often the strongest empirical results	Heuristic; many hyperparameters; no consistency guarantee

gives the noisy posterior

$$\boldsymbol{q}(x) := P(\tilde{Y} \mid x) = \boldsymbol{T}^{\top}\boldsymbol{p}^{\star}(x), \qquad \boldsymbol{p}^{\star}(x) = P(Y \mid x). \tag{1}$$

Both classifiers predict $\hat{y} = \arg\max_i p_i(x)$ at test time; $\boldsymbol{T}$ is used only during training and validation.

3.2 Classifier 1: forward loss correction

Forward correction [6] trains the network so that the induced noisy posterior $\boldsymbol{T}^{\top}\boldsymbol{p}(x)$ explains the noisy labels:

$$\mathcal{L}_{\mathrm{F}}(\theta) = -\frac{1}{N}\sum_{n=1}^{N} \log\left[\boldsymbol{T}^{\top}\boldsymbol{p}(x_n)\right]_{\tilde{y}_n} = -\frac{1}{N}\sum_{n=1}^{N} \log \sum_i p_i(x_n)\, T_{i\tilde{y}_n}. \tag{2}$$

Theory. The population version of (2) equals $\mathbb{E}_x \,\mathrm{KL}\big(\boldsymbol{q}(x) \,\|\, \boldsymbol{T}^{\top}\boldsymbol{p}(x)\big)$ plus a constant. It is minimised exactly when $\boldsymbol{T}^{\top}\boldsymbol{p}(x) = \boldsymbol{T}^{\top}\boldsymbol{p}^{\star}(x)$. If $\boldsymbol{T}$ is invertible, this forces $\boldsymbol{p} = \boldsymbol{p}^{\star}$, so the minimiser of the corrected *noisy* risk is the *clean* Bayes posterior, i.e. the method is consistent [6]. Forward correction never inverts $\boldsymbol{T}$, so unlike backward correction it cannot produce negative losses. **Optimisation.** We compute (2) in log space as $\mathrm{logsumexp}_i(\log p_i + \log T_{i\tilde{y}})$. This is numerically stable and also handles the exact zeros in $\boldsymbol{T}_{0.3}$. The objective is minimised with Adam (Sec. 3.5).

3.3 Classifier 2: generalised cross-entropy (GCE)

Formulation. For $q \in (0, 1]$, Zhang and Sabuncu [7] define the GCE loss

$$\mathcal{L}_q(\boldsymbol{p}(x), \tilde{y}) = \frac{1 - p_{\tilde{y}}(x)^q}{q}, \qquad \mathcal{L}_{\mathrm{GCE}}(\theta) = \frac{1}{N}\sum_{n=1}^{N} \mathcal{L}_q(\boldsymbol{p}(x_n), \tilde{y}_n). \tag{3}$$

It is trained directly on the noisy labels and does *not* use $\boldsymbol{T}$. We use the recommended $q = 0.7$.

Theory. As $q \to 0$, $\mathcal{L}_q$ becomes CE, and at $q = 1$ it equals $1 - p_{\tilde{y}}$, i.e. half the mean absolute error (MAE), a symmetric loss that is provably tolerant to symmetric label noise [10]. Its gradient is $\nabla_\theta \mathcal{L}_q = p_{\tilde{y}}^q \, (-\nabla_\theta \log p_{\tilde{y}})$: the CE gradient scaled by $p_{\tilde{y}}^q$. Examples that the model finds implausible, which are likely mislabelled, are therefore down-weighted. This is less aggressive than MAE (weight $p_{\tilde{y}}$), which underfits deep networks. The loss is bounded, $0 \le \mathcal{L}_q \le 1/q$, and satisfies $(C - C^{1-q})/q \le \sum_j \mathcal{L}_q(\boldsymbol{p}, j) \le (C-1)/q$, which yields noise-tolerance bounds that approach those of symmetric losses as $q \to 1$ [7]. Unlike forward correction, GCE has no consistency guarantee under asymmetric class-conditional noise. In exchange, it needs no estimate of $\boldsymbol{T}$ and is therefore unaffected by estimation errors.

Optimisation. $\mathcal{L}_{\mathrm{GCE}}$ is minimised with Adam (Sec. 3.5). The probability $p_{\tilde{y}}$ is clamped below at 10^{-12} before the power is taken, for numerical stability.

3.4 Transition-matrix estimator: anchor points

Principle. An anchor point of class i is an x^i with $P(Y{=}i \mid x^i) = 1$. By (1), $\boldsymbol{q}(x^i) = \boldsymbol{T}_{i,:}$, so the noisy posterior at an anchor reveals row i of $\boldsymbol{T}$ [5, 6]. If $\boldsymbol{T}$ is diagonally dominant, $q_i(x)$ increases with $P(Y{=}i \mid x)$, so anchors can be found as the maximisers of the noisy posterior. This is the estimator $\hat{T}_{ii} = \max_x \hat{P}(\tilde{Y}{=}i \mid x)$ from the lectures.

Algorithm. (1) On the 80% training partition of each split, train the network of Sec. 3.5 with plain CE on the noisy labels to obtain $\hat{\boldsymbol{q}}(x)$. Keep the checkpoint with the lowest CE on the 20% noisy validation partition (at most 20 epochs). (2) For each class i, take the top 1% of training images ranked by $\hat{q}_i(x)$ ($k = 144$ for FashionMNIST, $k = 120$ for CIFAR) as the approximate anchor set $\mathcal{A}_i$. (3) Set $\hat{\boldsymbol{T}}_{i,:} = \frac{1}{k} \sum_{x \in \mathcal{A}_i} \hat{\boldsymbol{q}}(x)$. Averaging over k points instead of using the single arg-max reduces variance and sensitivity to outliers. Each row is a convex combination of probability vectors, so $\hat{\boldsymbol{T}}$ is row-stochastic without any projection. Clean labels are never used.

Assumptions and biases. The estimator assumes that (near-)anchors exist, that $\hat{\boldsymbol{q}}$ is accurate, and that $\boldsymbol{T}$ is diagonally dominant, so that the clean and noisy class indices are aligned. Two opposite biases follow. An *over-confident* $\hat{\boldsymbol{q}}$ sharpens the anchor posteriors and *inflates* the diagonal. An *under-fitting* $\hat{\boldsymbol{q}}$ returns a mixture of rows at impure anchors and *shrinks* $\hat{\boldsymbol{T}}$ towards uniform. We also record two checks that need no clean labels. The first is the condition number of $\hat{\boldsymbol{T}}$, which governs the stability of the correction. The second is the column sums: if the clean classes are balanced, the noisy labels are balanced exactly when each column of $\boldsymbol{T}$ sums to one. For CIFAR, each split's $\hat{\boldsymbol{T}}$ is used by the forward-correction classifier trained on that split.

3.5 Network, pre-processing, optimisation and model selection

Network. All models are trained from scratch with one shared CNN. It has three blocks of [Conv3×3–BatchNorm–ReLU]×2 followed by max-pooling, with 32, 64 and 128 channels. These are followed by 3×3 average pooling, a 128-unit fully connected layer with dropout 0.25, and a 3-way output. No pretrained weights or external data are used.

Optimisation. We use Adam [12] with learning rate 10^{-3}, weight decay 10^{-4}, cosine annealing, batch size 128 and at most 30 epochs, in PyTorch with deterministic kernels and fixed seeds.

Pre-processing. Pixels are only divided by 255. FashionMNIST vectors are reshaped to $1\times28\times28$ and CIFAR images to channel-first $3\times32\times32$. We deliberately use no standardisation statistics and no augmentation. Nothing is fitted on validation or test data, every method sees identical inputs, and the effect of the noise-handling method is not confounded with augmentation.

Model selection and over-fitting control. No clean labels are available for validation, so each method is validated with its own objective on the 20% noisy hold-out after every epoch. Forward correction uses the forward-corrected NLL $-\log[\boldsymbol{T}^\top \boldsymbol{p}(x)]_{\tilde{y}}$, whose population minimiser is the clean posterior (see above). GCE uses its own loss (3) on the noisy validation labels, so it remains completely matrix-free. The best checkpoint is restored, and training stops early after 8 epochs without improvement. This early stopping also counters memorisation [3]. The hyper-parameters (q, learning rate, epoch budget, anchor fraction) were fixed in advance from the literature and pilot runs on noisy validation data. None was tuned on test accuracy. The clean test set is read only after the checkpoint has been selected.

4 Experiments

4.1 Setup

Datasets. FashionMNIST0.3 and FashionMNIST0.6 [13] each contain $n = 18{,}000$ noisy 28×28 grey-scale images, and CIFAR [14] contains $n = 15{,}000$ noisy 32×32×3 colour images. Each dataset has $m = 3{,}000$ test images. The noisy labels are nearly balanced (5918–6112 per class for FashionMNIST, exactly 5000 for CIFAR). The given matrices are $\boldsymbol{T}_{0.3}$ (rows $[.7, .3, 0]$, $[0, .7, .3]$, $[.3, 0, .7]$; condition number 1.64) and $\boldsymbol{T}_{0.6}$ (diagonal 0.4, off-diagonal 0.3; condition number 10.0). The CIFAR matrix is unknown. Test sets are clean and balanced (1000 per class).

Protocol. For each dataset we draw ten independent uniformly random 80/20 partitions (seeds 0–9). This gives 14,400/3,600 examples for FashionMNIST and 12,000/3,000 for CIFAR. On each split we run the estimator once and train both classifiers from the same initialisation and batch order, so the comparison is *paired*. The 60 classifier runs and 30 estimator runs were executed in one session on a single NVIDIA A100-SXM4-80GB (PyTorch 2.15, CUDA 13.2) and took about 18 minutes (7–19 s per classifier run). Forward correction uses the given $\boldsymbol{T}$ on FashionMNIST and each split's $\hat{\boldsymbol{T}}$ on CIFAR; GCE never uses $\boldsymbol{T}$.

Table 2: Clean test top-1 accuracy (%), mean ± SD over 10 random splits (FMNIST = FashionMNIST). Δ: paired GCE − Forward (percentage points); p: two-sided paired t-test / Wilcoxon test.

Dataset	Forward	GCE	Δ (pp)	p (t / W)
FMNIST0.3 (given $\boldsymbol{T}$)	$\mathbf{99.18 \pm 0.18}$	98.66 ± 0.33	-0.52 ± 0.29	0.0003 / 0.002
FMNIST0.6 (given $\boldsymbol{T}$)	$\mathbf{92.49 \pm 2.01}$	91.44 ± 4.31	-1.06 ± 4.82	0.51 / 0.63
CIFAR (estimated $\hat{\boldsymbol{T}}$)	$\mathbf{66.23 \pm 15.98}$	64.89 ± 5.34	-1.33 ± 16.22	0.80 / 0.23

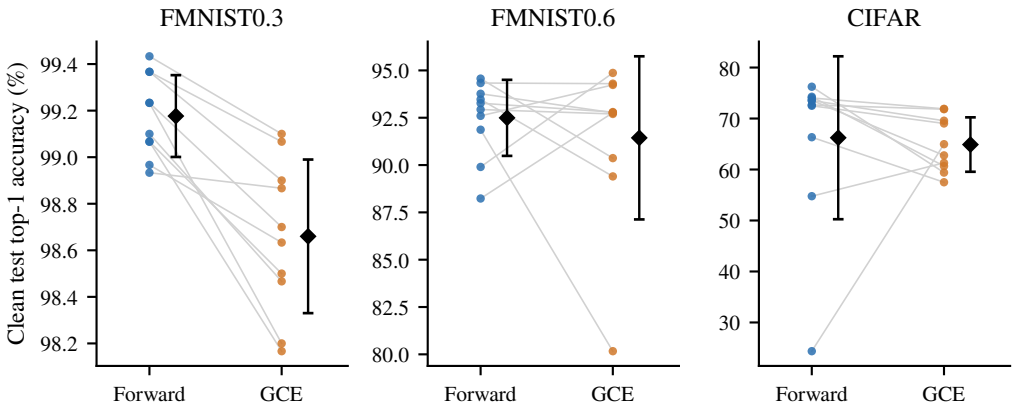


Figure 1: Per-split clean test accuracy. Grey lines connect the two classifiers trained on the same split; diamonds and bars show the mean ± SD.

Metric. Clean test top-1 accuracy, reported as mean ± sample SD over the ten splits, with paired t-tests and Wilcoxon signed-rank tests on the per-split differences.

4.2 Classification with known flip rates (FashionMNIST)

Table 2 and Figure 1 report the results. At a 30% flip rate both classifiers almost reach the noise-free regime. Forward correction ($99.18 \pm 0.18\%$) beats GCE ($98.66 \pm 0.33\%$) on all ten splits; the gap is small but significant ($p = 0.0003$, paired t-test). The remaining errors are mostly confusions between classes 0 and 2, which GCE makes about twice as often for class 2 (recall 97.3% vs. 98.8%). This is consistent with theory: $\boldsymbol{T}_{0.3}$ is strongly asymmetric, so the knowledge of $\boldsymbol{T}$ that forward correction exploits is most valuable here, whereas GCE's tolerance guarantee covers only symmetric noise.

The near-uniform $\boldsymbol{T}_{0.6}$ keeps only 40% of labels correct. It costs Forward 6.7 points and GCE 7.2 points, and the variability grows by an order of magnitude (SD $0.18 \to 2.01$ and $0.33 \to 4.31$). Forward is again more accurate on average and has a better worst case (88.2% vs. 80.2%), but the paired difference is not significant ($p = 0.51$) and GCE wins on 3 of the 10 splits. Near-symmetric noise is the regime in which a matrix-free robust loss is expected to be competitive.

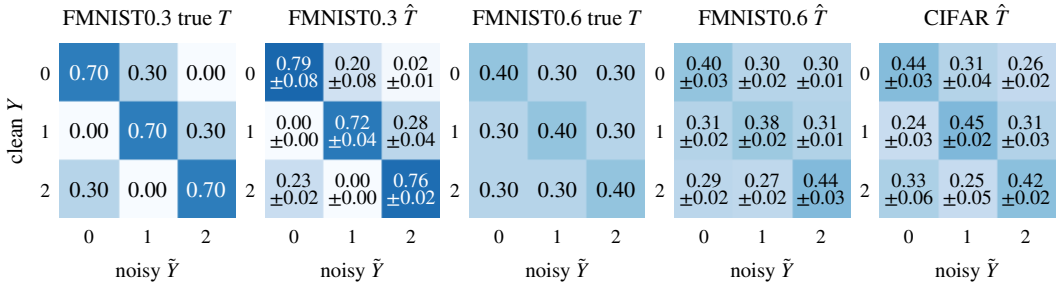


Figure 2: True and estimated transition matrices (estimates: mean ± SD over 10 splits). Rows are clean classes and columns are noisy labels.

Table 3: Estimator diagnostics over the 10 splits. MAE = mean absolute entry-wise error and $\|\cdot\|_F$ = Frobenius error with respect to the given $\boldsymbol{T}$ (mean ± SD).

	FMNIST0.3	FMNIST0.6	CIFAR
MAE	0.045 ± 0.020	0.022 ± 0.008	— ($\boldsymbol{T}$ unknown)
$\|\hat{\boldsymbol{T}} - \boldsymbol{T}\|_F$	0.188 ± 0.088	0.082 ± 0.032	—
max. entry SD	0.082	0.034	0.057
cond($\hat{\boldsymbol{T}}$) range	1.54–1.80	8.52–27.61	6.30–10.04
column sums	1.02 / 0.92 / 1.06	0.99 / 0.95 / 1.06	1.01 / 1.00 / 0.99
estimator noisy val. acc.	68–71%	37–40%	37–40%

4.3 Transition-matrix estimation

Validation on known matrices (Figure 2, Table 3). We compare $\hat{\boldsymbol{T}}$ with the given matrices entry-wise (MAE and Frobenius error) and structurally. On FashionMNIST0.3 the estimator recovers the structure of $\boldsymbol{T}$: the cyclic flips $0\to1$, $1\to2$, $2\to0$ are found, and the structural zeros are estimated as ≤ 0.02. The MAE is 0.045 ± 0.020. The error is systematic. The diagonal is over-estimated (0.72–0.79 instead of 0.70), and the flip probabilities are under-estimated (0.20–0.28 instead of 0.30), most strongly for class 0 on a single split (MAE 0.086). This is the over-confidence bias of Sec. 3.4: FashionMNIST is easy, so the CE network becomes very confident on its most typical images, and its posteriors there are sharper than the true rows [15]. On FashionMNIST0.6 the MAE is 0.022 ± 0.008, and in every row the diagonal is correctly ranked above the off-diagonal entries. The condition number of $\hat{\boldsymbol{T}}$, however, varies between 8.5 and 27.6 (true value 10.0), which shows how sensitive near-uniform matrices are to small errors.

Supplementary estimator study. A group member also tested a *pseudo-confusion* estimator under a separate protocol, so its numbers are not directly comparable. It tabulates held-out noisy labels within arg-max pseudo-classes and reached an MAE of 0.013 (FashionMNIST0.3) and 0.015 (FashionMNIST0.6). On CIFAR, however, its matrix gave lower Forward accuracy in 7 of 10 paired splits, so we kept the anchor estimator.

4.4 Classification with unknown flip rates (CIFAR)

Pre-processing and formulation. CIFAR uses the same pre-processing, network, optimiser and model-selection rule as FashionMNIST (Sec. 3.5). The only difference is that forward correction replaces the given $\boldsymbol{T}$ by the split's estimate $\hat{\boldsymbol{T}}$ in (2) and in its validation criterion. GCE is unchanged, since it never uses $\boldsymbol{T}$.

Estimated transition matrix. Averaged over the ten splits (mean $\pm$ SD; each Forward classifier used its own split's estimate), our estimate is

$$\hat{\boldsymbol{T}}_{\mathrm{CIFAR}} = \begin{bmatrix} 0.436 \pm 0.032 & 0.306 \pm 0.037 & 0.258 \pm 0.024 \\ 0.241 \pm 0.031 & 0.450 \pm 0.018 & 0.309 \pm 0.034 \\ 0.332 \pm 0.057 & 0.249 \pm 0.049 & 0.419 \pm 0.022 \end{bmatrix}. \tag{4}$$

The estimated matrix suggests heavy, asymmetric label noise, with a mean diagonal near 0.43. Since the true CIFAR transition matrix is unavailable, this is an estimate rather than a measured clean-label proportion. The dominant flips $0\to1$ (0.31), $1\to2$ (0.31) and $2\to0$ (0.33) follow the same cyclic pattern as $\boldsymbol{T}_{0.3}$, which suggests a mixture of a cyclic and a symmetric component. Two checks that need no clean labels assess the estimate. The column sums (1.01 / 1.00 / 0.99) agree with the exactly balanced noisy labels, and $\hat{\boldsymbol{T}}$ is well enough conditioned (6.3–10.0) for correction. Because the CE network reaches only 37–40% noisy validation accuracy, its anchors are impure. The under-fitting bias discussed in Sec. 3.4 may shrink $\hat{\boldsymbol{T}}$ towards uniformity. Without clean labels, however, neither the magnitude nor direction of this bias can be verified.

Accuracy. Although the estimated transition matrix suggests substantial noise, both methods generalise above chance on clean CIFAR test data (Table 2). Forward correction with $\hat{\boldsymbol{T}}$ reaches $66.23 \pm 15.98\%$ and GCE $64.89 \pm 5.34\%$. The means hide two very different behaviours. Forward correction is the better method on 8 of the 10 splits (66.3–76.2%, $72.9 \pm 2.9\%$; median 73.1% vs. 63.9% for GCE) but fails badly on two. On split 1 it reaches only 24.4%, below chance: its predictions are a *cyclic permutation* of the classes (86% of class-2 images are predicted as class 0, 63% of class-1 images as class 2). On split 7, class 0 is almost entirely predicted as class 1 (recall 3.1%). GCE never fails in this way (57.5–71.9%) and wins exactly on these two splits, so the paired difference is not significant ($p = 0.80$). The measured estimator diagnostics alone do not explain the failures: across splits, accuracy correlates only weakly with the condition number or mean diagonal of $\hat{\boldsymbol{T}}$ (Spearman $|\rho| \leq 0.44$), and split 7's estimate is typical. We analyse their cause in Sec. 4.5.

4.5 Discussion

Model selection under heavy noise. Even a perfect classifier cannot push the validation forward NLL below the noise entropy $\mathbb{E}_Y H(\boldsymbol{T}_{Y,:})$. For $\boldsymbol{T}_{0.6}$ this floor is 1.089 nats, only 0.010 below the chance level $\ln 3 = 1.099$. For CIFAR under $\hat{\boldsymbol{T}}$ it is 1.072 nats, a window of 0.027. Checkpoint selection for forward correction therefore happens inside a very narrow window (Figure 3), where the sampling noise of a 3,000-example validation set is comparable to the differences between check-

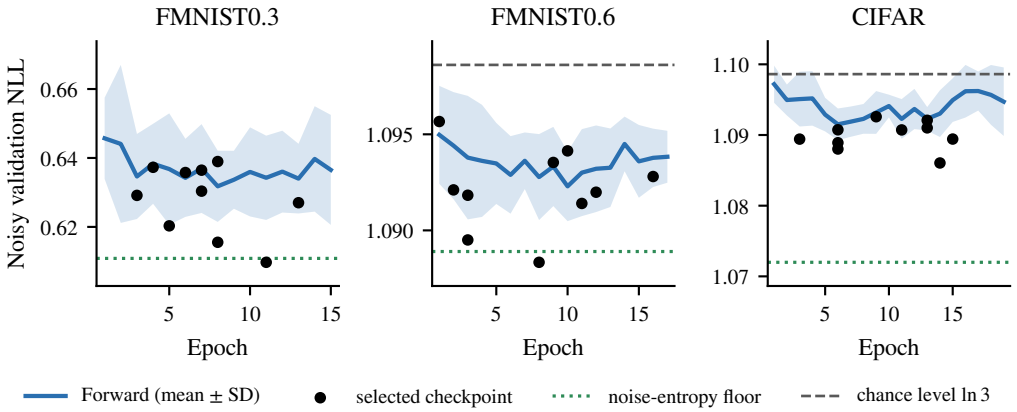


Figure 3: Forward-corrected NLL of the Forward classifier on the noisy validation set: mean $\pm$ SD over the splits still training (line and band) and the checkpoint selected in each split (dots). Dotted: population noise-entropy floor $\mathbb{E}_Y H(\boldsymbol{T}_{Y,:})$ (CIFAR: under $\hat{\boldsymbol{T}}$); dashed: chance level $\ln 3$, which lies above the FMNIST0.3 panel. Finite-sample values can fall slightly below the floor.

points, so a slightly different selected epoch can mean a large difference in clean accuracy. This may contribute to the large SDs of forward correction on Fashion-MNIST0.6 and CIFAR. The CIFAR failures are consistent with this difficulty: the checkpoint selected on split 1, which permutes the classes, has a noisy validation NLL of 1.0926, barely outside the range of the successful splits (1.086–1.092). Under such heavy, nearly circulant noise, the noisy likelihood hardly distinguishes the correct class order from a permuted one, so neither training nor model selection reliably rejects the permuted solution.

Comparing the two classifiers. The two methods handle the noise in different ways. Forward correction models the noise explicitly through $\boldsymbol{T}$ and is consistent when $\boldsymbol{T}$ is correct, so its accuracy depends on the quality of $\boldsymbol{T}$ or $\hat{\boldsymbol{T}}$. GCE ignores $\boldsymbol{T}$ and relies on its bounded loss, which down-weights examples whose label the model finds implausible. It is unaffected by estimation errors but has no guarantee under asymmetric noise. The results match this trade-off. With the strongly asymmetric $\boldsymbol{T}_{0.3}$, knowing $\boldsymbol{T}$ gives forward correction a consistent, significant edge. With the near-symmetric $\boldsymbol{T}_{0.6}$, the two are statistically indistinguishable. On CIFAR, forward correction is clearly more accurate when it identifies the classes correctly. GCE is much more stable (SD 5.3 vs. 16.0), however, because it neither depends on $\hat{\boldsymbol{T}}$ nor needs the noisy likelihood to resolve the class order.

Robustness and reflection. Neither method is uniformly more robust. Forward correction has the higher mean accuracy on every dataset and degrades slightly less from 30% to 60% noise (6.7 vs. 7.2 points). GCE is the more reliable choice when $\boldsymbol{T}$ must be estimated under heavy noise: it has no catastrophic failures and a three times smaller SD on CIFAR. In practice, the two could be combined, e.g. by using GCE predictions to check that the forward-corrected model has not permuted the classes. A natural refinement is a validation criterion that estimates the *clean* risk

directly, e.g. an importance-weighted validation error [5], which would put both methods on the same footing. The near-perfect accuracy expected at a 30% flip rate leaves little room to separate the methods, so the informative comparisons are the high-noise ones.

5 Conclusion and future work

We studied classification under class-conditional label noise with forward correction, which uses the transition matrix, and the matrix-free GCE loss, together with an anchor-point estimator of $\boldsymbol{T}$. With the given matrices, forward correction is more accurate than GCE at a 30% flip rate (99.2% vs. 98.7%, significant) and comparable at a 60% flip rate (92.5% vs. 91.4%). The estimator recovers the known matrices with MAE 0.045 and 0.022, and its systematic errors are explained by over-confident posteriors. On CIFAR the estimated matrix has a diagonal near 0.43; the true transition matrix is unavailable. With this estimate, forward correction reaches 73% on most splits but converges to a class-permuted solution on two, giving $66.2 \pm 16.0\%$, while GCE is lower but far more stable ($64.9 \pm 5.3\%$). Under heavy noise, model selection on noisy validation data operates in a very narrow likelihood window, which makes results variable across splits. Future work: (i) select models with an importance-weighted estimate of the clean risk; (ii) reduce the estimator's bias with temperature calibration [15], Dual-T [8] or T-revision [9]; (iii) use augmentation and deeper networks on CIFAR to obtain purer anchors and lower variance; (iv) combine GCE with $\boldsymbol{T}$, e.g. through importance reweighting [5], to test whether the two ideas are complementary.

A How to run the code

Environment. Python $\geq$3.10 with the packages in `requirements.txt` (PyTorch $\geq$2.5, NumPy, Pillow). The reported results were produced on one NVIDIA A100-SXM4-80GB with Python 3.11, PyTorch 2.15.0 and CUDA 13.2. **Data.** Place `FashionMNIST0.3.npz`, `FashionMNIST0.6.npz` and `CIFAR.npz` directly in the empty `data/` folder (or add `--data-dir <path>` to the commands). **Commands**, run from the root of the code folder:

```
python -m pip install -r requirements.txt
python -m unittest discover -s algorithm/tests
python -m algorithm.run_experiments \
    --config algorithm/configs/final.json --device cuda
python -m algorithm.summarize \
    --results-dir results/final_standard_gce
```

These commands install dependencies, run the tests and execute the full 3-dataset, 10-split estimator/Forward/GCE grid. Summary: `results/final_standard_gce/summary.md`. Per-split files: `estimator/transition.json`, `forward/metrics.json`, `gce/metrics.json`. Use `--resume` to continue interrupted runs; configurations are in `algorithm/configs/final.json`.

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